Solution

In triangle ABC, consider A as the origin. Then the position vectors of D, E, F are given by $\frac{\overrightarrow{AB}+\overrightarrow{AC}}{2}, \frac{\overrightarrow{AC}}{2}, \frac{\overrightarrow{AB}}{2}$ respectively. Since $|\overrightarrow{AB}\times\overrightarrow{AC}|$ is the area of the parallelogram formed by the two vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ as adjacent sides, the area of ΔABC is $\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$. Similarly, considering ΔDEF, we have

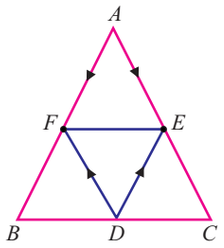


Fig. 6.13

$$\begin{aligned}
\text{the area of } \Delta DEF &= \frac{1}{2}|\overrightarrow{DE}\times\overrightarrow{DF}| \\
&= \frac{1}{2}|(\overrightarrow{AE}-\overrightarrow{AD})\times(\overrightarrow{AF}-\overrightarrow{AD})| \\
&= \frac{1}{2}\left|\frac{\overrightarrow{AB}}{2}\times\frac{\overrightarrow{AC}}{2}\right| \\
&= \frac{1}{4}\left(\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|\right) \\
&= \frac{1}{4}\ (\text{the area of } \Delta ABC)\,.
\end{aligned}$$

6.3.4 Application of dot and cross product in Physics

Definition 6.2

If $\vec{d}$ is the displacement vector of a particle moved from a point to another point after applying a constant force $\vec{F}$ on the particle, then the **work done by the force** on the particle is $w=\vec{F}\cdot\vec{d}$.

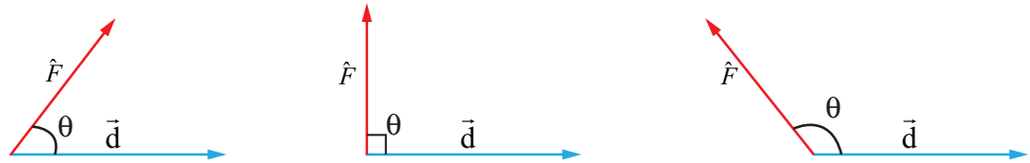


Fig. 6.14

If the force has an acute angle, perpendicular angle, and an obtuse angle, the work done by the force is positive, zero, and negative respectively.

Example 6.9

A particle acted upon by constant forces $2\hat{i}+5\hat{j}+6\hat{k}$ and $-\hat{i}-2\hat{j}-\hat{k}$ is displaced from the point $(4,-3,-2)$ to the point $(6,1,-3)$. Find the total work done by the forces.

Solution

Resultant of the given forces is $\vec{F}=(2\hat{i}+5\hat{j}+6\hat{k})+(-\hat{i}-2\hat{j}-\hat{k})=\hat{i}+3\hat{j}+5\hat{k}$.

Let A and B be the points $(4,-3,-2)$ and $(6,1,-3)$ respectively. Then the displacement vector of the particle is $\vec{d}=\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=(6\hat{i}+\hat{j}-3\hat{k})-(4\hat{i}-3\hat{j}-2\hat{k})=2\hat{i}+4\hat{j}-\hat{k}$.

Therefore the work done $w=\vec{F}\cdot\vec{d}=(\hat{i}+3\hat{j}+5\hat{k})\cdot(2\hat{i}+4\hat{j}-\hat{k})=9$ units.

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